ODV 4. DOMAĆA

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$$f(x_1, x_2, x_3, x_4) = (x_1 \uparrow x_2) \rightarrow (x_3 \downarrow x_4)$$

1) TABELA

x_1	x_2	x_3	x_4	$f(x_1\, x_2\, x_3\, x_4)$	
0	0	0	0	1	T_0
0	0	0	1	0	
0	0	1	0	0	
0	0	1	1	0	
0	1	0	0	1	M
0	1	0	1	0	M
0	1	1	0	0	
0	1	1	1	0	S
1	0	0	0	1	S
1	0	0	1	0	
1	0	1	0	0	
1	0	1	1	0	
1	1	0	0	1	
1	1	0	1	1	
1	1	1	0	1	
1	1	1	1	1	T_1

2) PRIPADNOST RAZREDOM

NABOR = $\{1, 0\}$

T_0 OHRANJA NIČLE

f NE PRIPADA RAZREDU T_0; $f \notin T_0$

OZNAČENO V TABELI

T_1 OHRANJA ENICE

f PRIPADA RAZREDU; $f \in T_1$

OZNAČENO V TABELI

PREVERIMO ŠE ZA $\{0\}$

0 NE PRIPADA T_1;

$0 \notin T_1$

S SEBIDUALNE FUNKCIJE

f NE PRIPADA RAZREDU; $f \notin S$

OZNAČENO V TABELI

L LINEARNE FUNKCIJE

$$f(x_1, x_2, x_3, x_4) = a_0 \nabla x_1 a_1 \nabla x_2 a_2 \nabla x_3 a_3 \nabla x_4 a_4$$

$f(0,0,0,0) = \boxed{a_0} \nabla a_1 0 \nabla a_2 0 \nabla a_3 0 \nabla a_4 0 = 1 \qquad (a_0 = 1)$

$f(1,0,0,0) = 1 \nabla \boxed{a_1 1} \nabla a_2 0 \nabla a_3 0 \nabla a_4 0 = 1 \qquad (a_1 = 0)$

$f(0,1,0,0) = 1 \nabla 0 \cdot 0 \nabla \boxed{a_2 1} \nabla a_3 0 \nabla a_4 0 = 0 \qquad (a_2 = 1)$

$f(0,0,1,0) = 1 \nabla 0 \cdot 0 \nabla 1 \cdot 0 \nabla \boxed{a_3 1} \nabla a_4 0 = 0 \qquad (a_3 = 1)$

$f(0,0,0,1) = 1 \nabla 0 \cdot 0 \nabla 1 \cdot 0 \nabla 1 \cdot 0 \nabla \boxed{a_4 \cdot 1} = 1 \qquad (a_4 = 0)$

$f(x_1 x_2 x_3 x_4)_L = 1 \nabla x_2 \nabla x_3$

$f(x_1 x_2 x_3 x_4)_L = f(x_1 x_2 x_3 x_4)$

$f(0,1,0,0)_L = f(0,1,0,0)$

$1 \nabla 1 \nabla 0 = 0 \qquad = 1$ (IZ TABELE)

$0 \neq 1$ NI LINEARNA

$f \notin L$

M MONOTONE FUNKCIJE

$$f(0,1,0,0) \leq f(0,1,0,1)$$

$$1 \leq 0 \quad //$$

$f(\,)$ NE PRIPADA MONOTONIM; $f \notin M$

ZAKLJUČEK

FUNKCIJI f in 0 ODPIRATA VSE OSNOVNE RAZREDE $\Longrightarrow$ $\{f, 0\}$ je poln nabor.